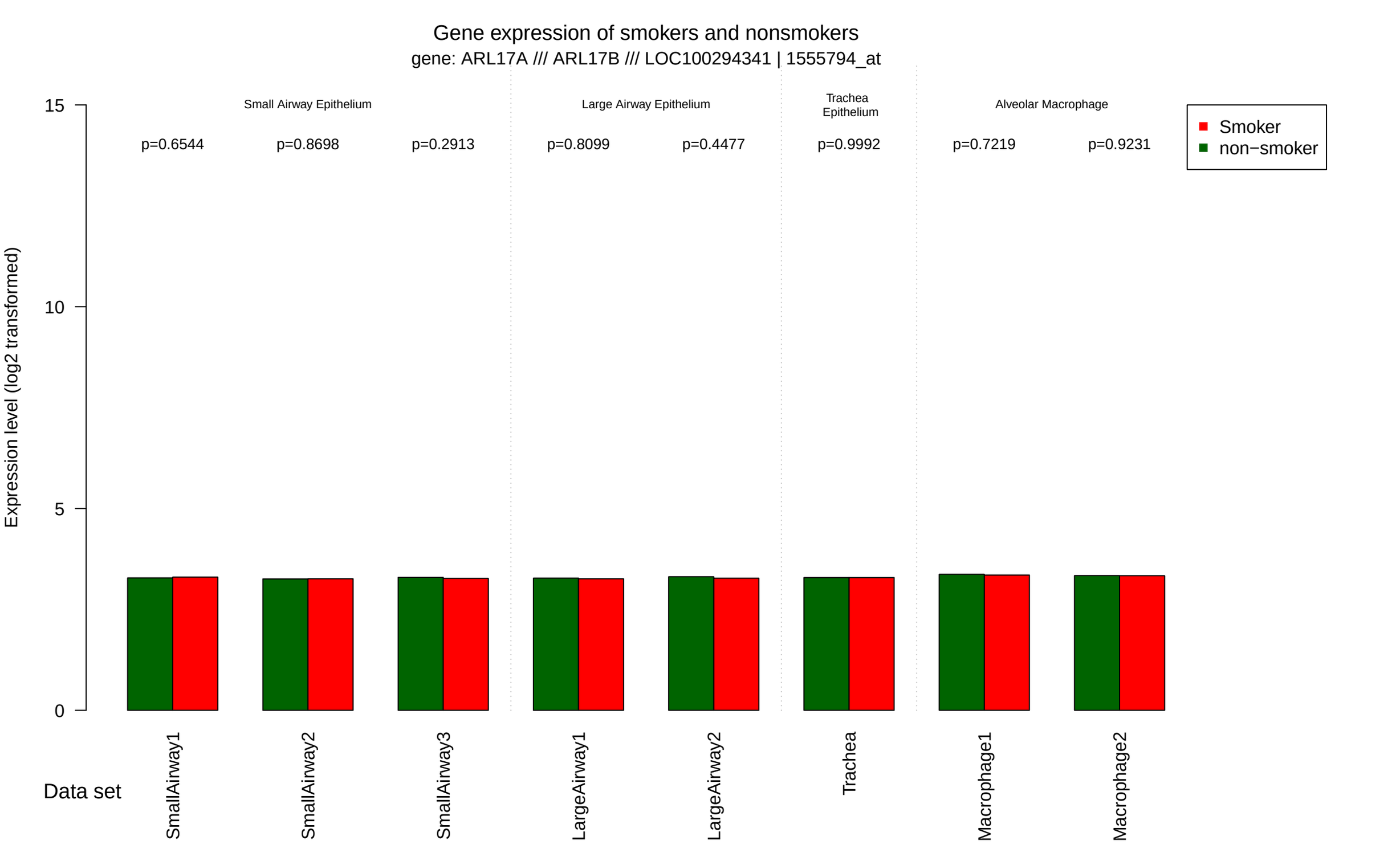
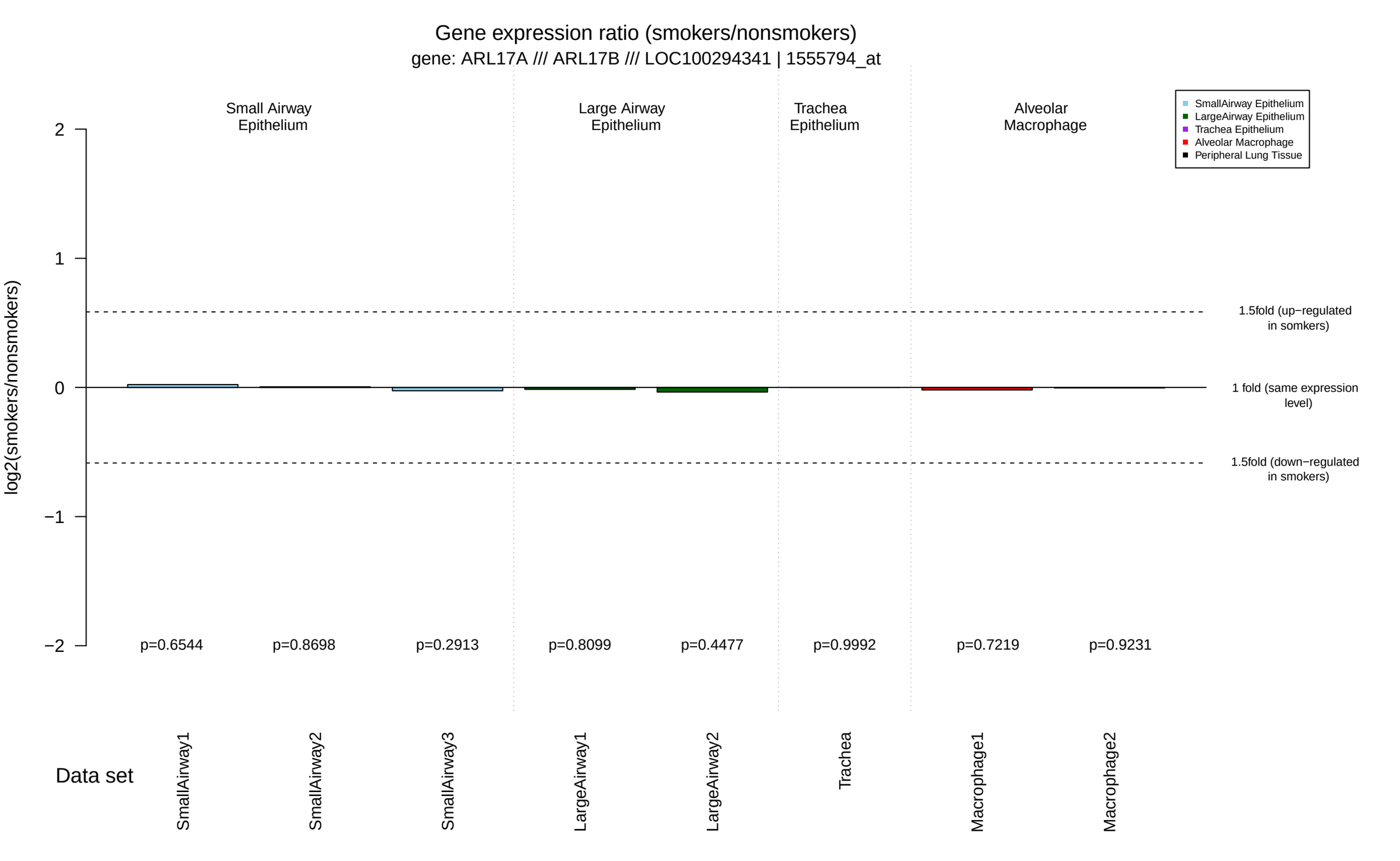
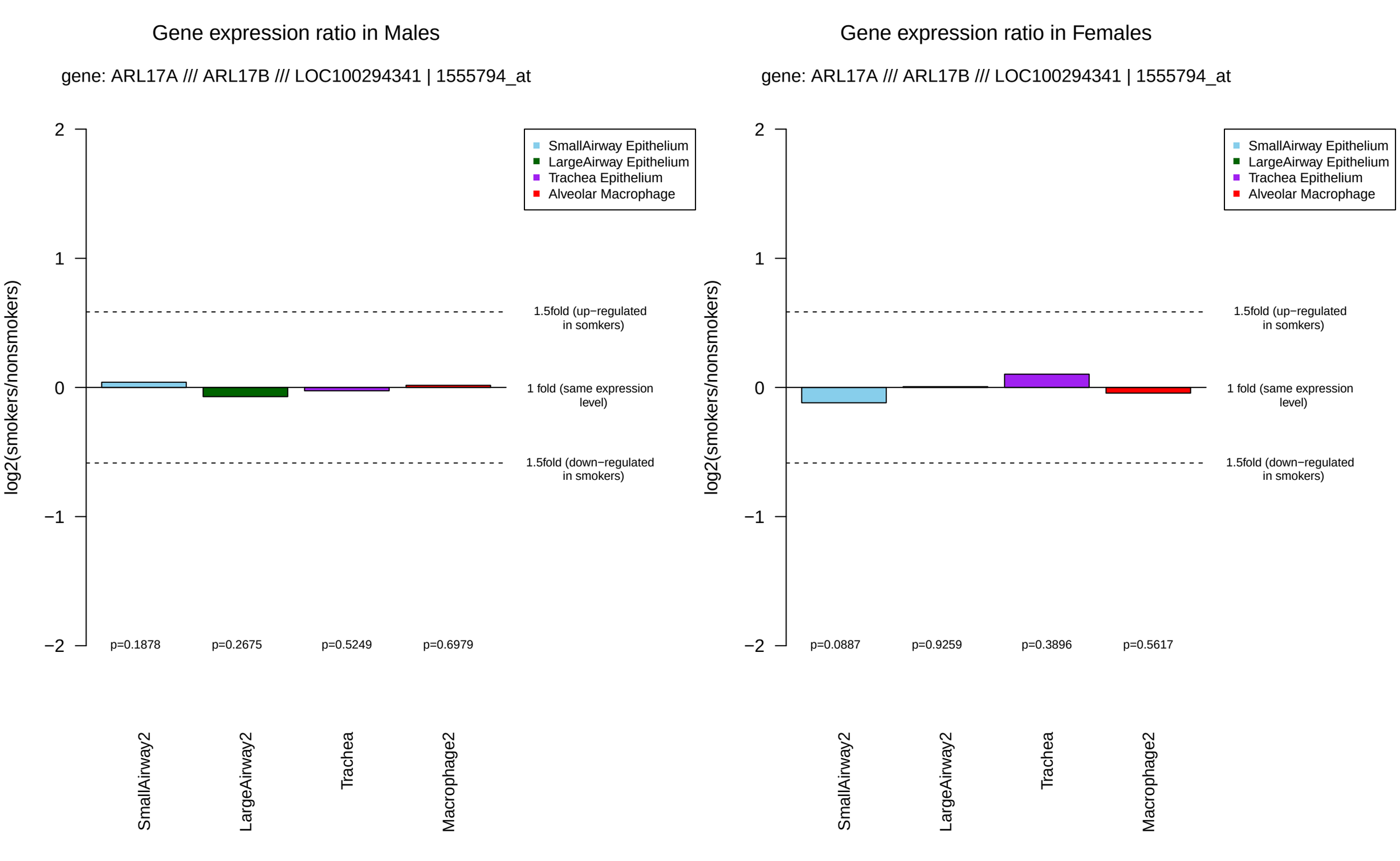
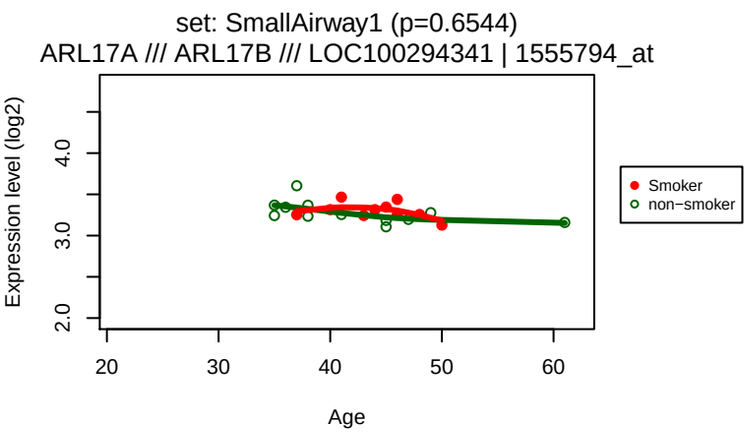
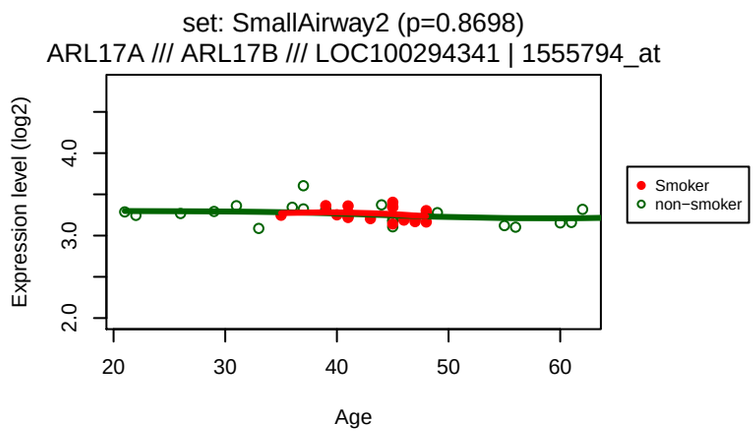
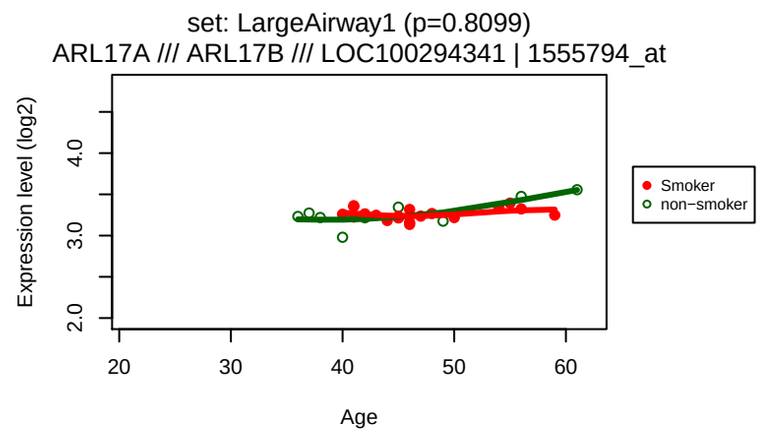

Gene expression of smokers and nonsmokers
gene: ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6544
p=0.8698
p=0.2913
p=0.8099
p=0.4477
p=0.9992
p=0.7219
p=0.9231
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

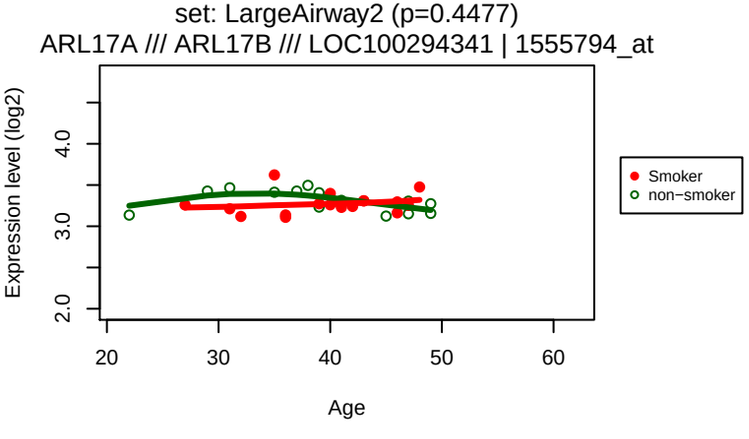

Gene expression ratio (smokers/nonsmokers)
gene: ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6544
p=0.8698
p=0.2913
p=0.8099
p=0.4477
p=0.9992
p=0.7219
p=0.9231
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

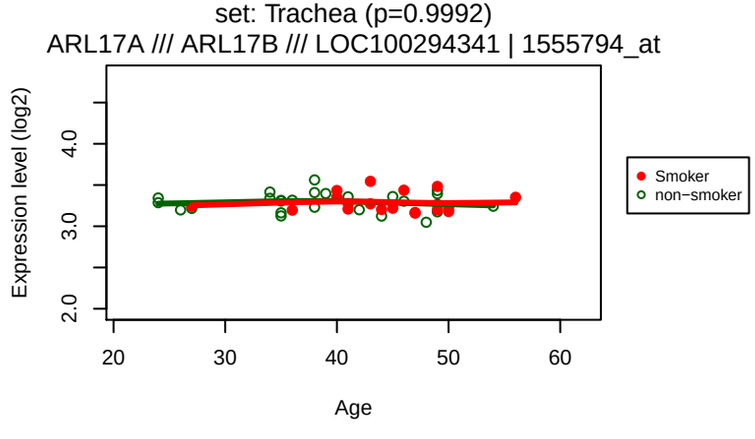

Gene expression ratio in Males
gene: ARL17A /// ARL17B /// LOC100294341 | 1555794_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1878
p=0.2675
p=0.5249
p=0.6979
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL17A /// ARL17B /// LOC100294341 | 1555794_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0887
p=0.9259
p=0.3896
p=0.5617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

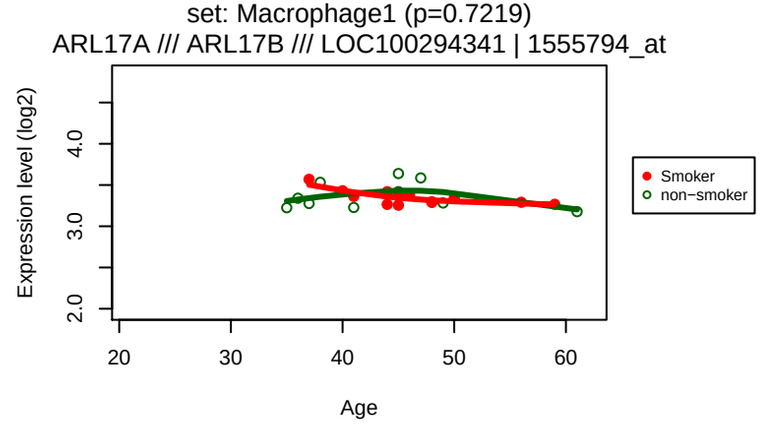

set: SmallAirway1 (p=0.6544)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

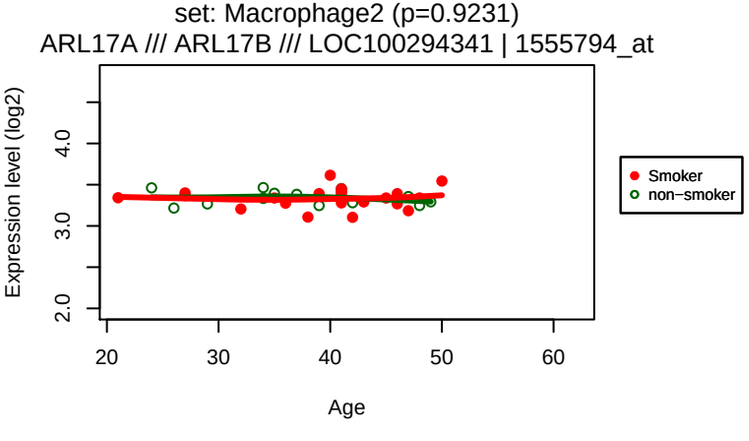

set: SmallAirway2 (p=0.8698)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker

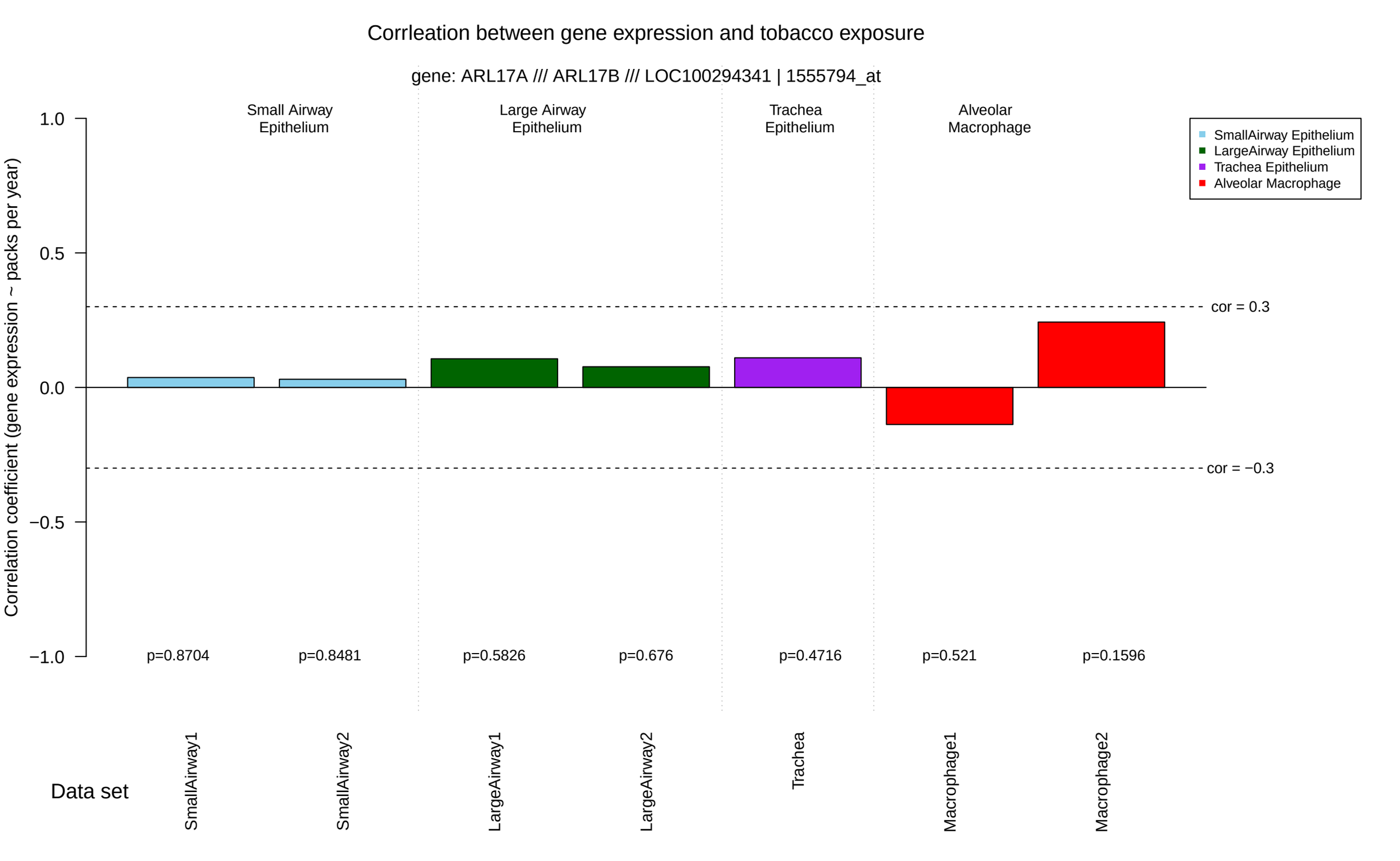

set: LargeAirway1 (p=0.8099)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4477)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9992)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7219)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9231)
ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Expression level (log2)
2.0
3.0
4.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL17A /// ARL17B /// LOC100294341 | 1555794_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8704
p=0.8481
p=0.5826
p=0.676
p=0.4716
p=0.521
p=0.1596
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2